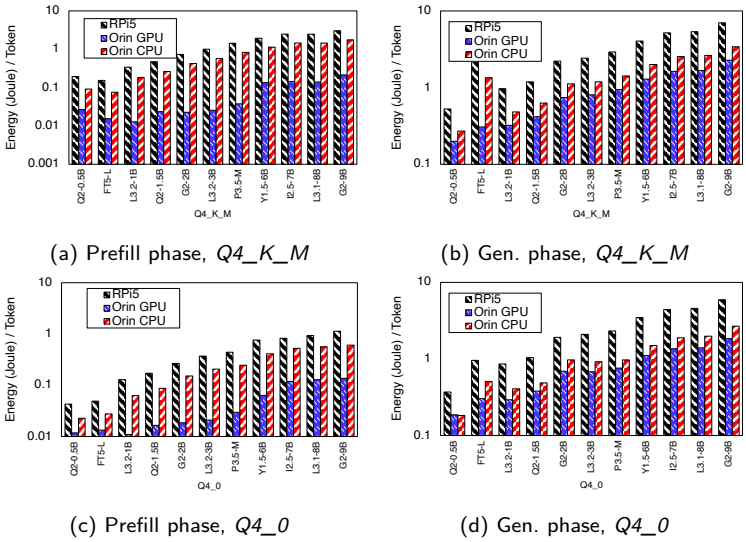


(a) Prefill phase, *Q4_K_M* (b) Gen. phase, *Q4_K_M*

(c) Prefill phase, *Q4_0* (d) Gen. phase, *Q4_0*

Fig. 10. Energy per token for all models, quantization schemes, and all devices

up to 20% during generation phase. This reduction is not solely due to increased inference speed but also to a moderate decrease in power draw compared to *Q4_K_M* quantization.

When examining model-specific energy consumption in Figure 10, a general trend of increasing energy usage with model size is evident. This is primarily driven by longer inference latencies, although a slight increase in power draw is also observed. However, for Phi 3.5 and all larger models, average power draw remained approximately constant. In the generation phase, Flan T5 Large emerged as a notable outlier, exhibiting substantially higher energy consumption relative to its size. This inefficiency stems from its lower generation speed, leading to longer active power draw.

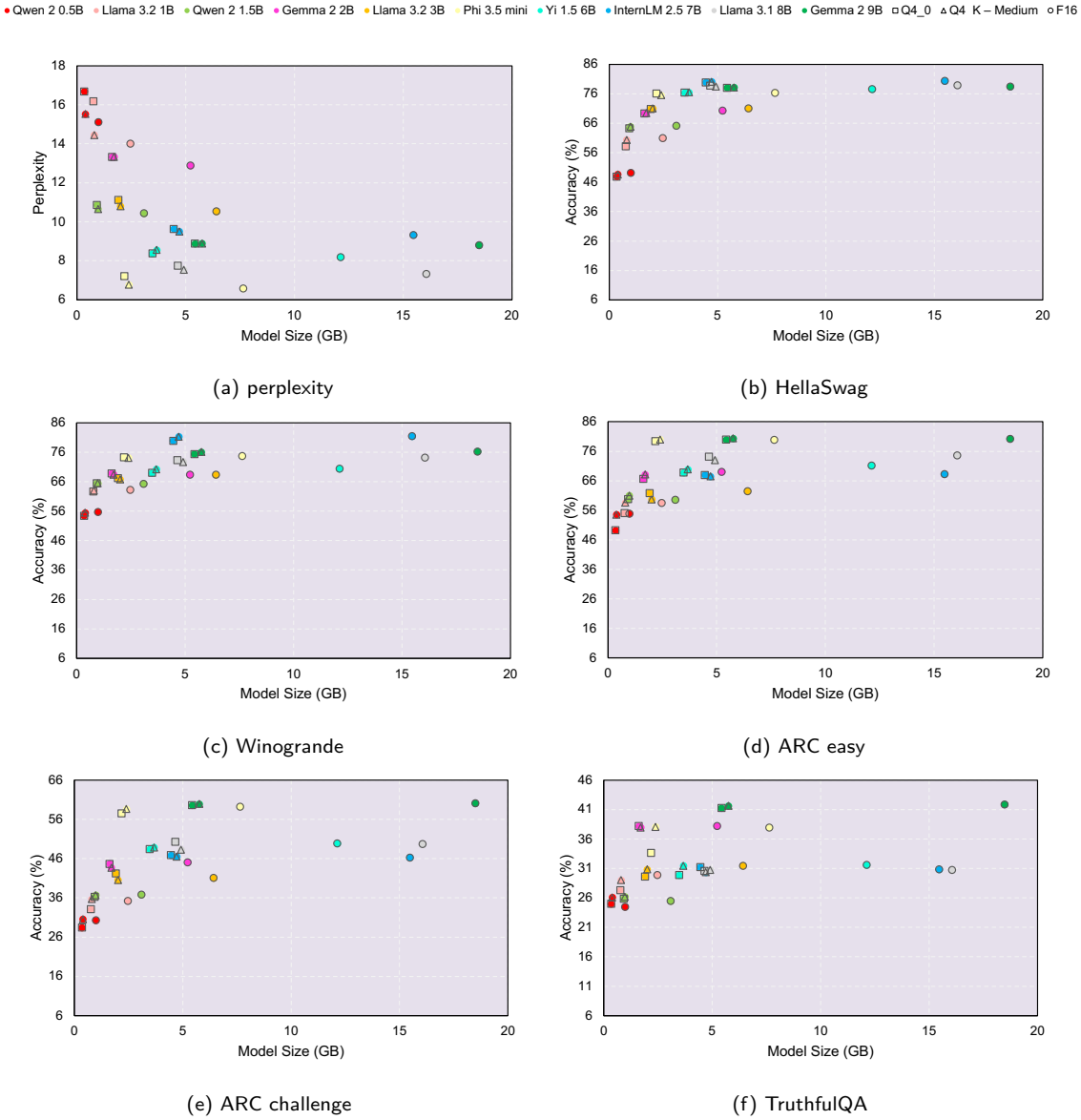
For all models, GPU inference significantly reduced energy consumption compared to the RPi 5, with improvements by factors of ~ 3.6–~ 39.3 in the prefill phase and ~ 1.9–~ 7.8 in the generation phase, depending on the model and quantization scheme. For CPU inference, the Orin also demonstrated superior energy efficiency over the RPi 5 in both phases. In the prefill phase, this improvement was primarily due to a lower mean power draw, resulting in energy reductions of ~ 36.0–$\sim 52.5\%$. In the generation phase, the lower inference latencies on the Orin further amplified this difference, leading to energy consumption reductions of ~ 47–$\sim 58\%$. *Q4_0* quantization consistently outperformed *Q4_K_M* quantization in terms of energy efficiency for almost all models and devices, mainly due to reduced inference time but also because of a slightly lower power draw. In the prefill phase, the efficiency improvement factors ranged between ~ 1.1 and ~ 4.5, depending on the model and device. In the generation phase, improvements were ~ 1.008 to ~ 2.6.

Main Takeaways

- GPU inference is more energy efficient than CPU inference in both prefill and generation phase

- *Q4_0* quantization improves energy efficiency on all devices for both prefill and generation phase over *Q4_K_M* quantization

5.4 Quantization Effect



(a) perplexity (b) HellaSwag

(c) Winogrande (d) ARC easy

(e) ARC challenge (f) TruthfulQA

Fig. 11. Results of qualitative benchmarks

Since quantized models were used in this evaluation, it is essential to assess how quantization impacts both output quality and computational performance compared to unquantized models. To analyze the effects

on output quality, a a suite of five benchmarks were conducted, with results presented in Figure 11. The findings reveal a perplexity increase across both quantization schemes and all models when compared to their 16-bit floating point counterparts. This effect was particularly pronounced for the smallest models. Additionally, a clear distinction between quantization schemes emerged: *Q4_0* quantization exhibited higher perplexity than *Q4_K_M* quantization for all models, except for Yi 1.5 6B and the Gemma models, where both schemes performed similarly. However, on the down-stream task benchmarks the observed accuracy drop was almost in all cases negligible, showing the effectiveness of quantization as a model compression technique. These results go hand in hand with previous work which has shown that 4-bit quantization of LMs impacts model quality on downstream tasks only modestly, while perplexity was found to be a more sensitive metric for evaluating quantization effects [12].

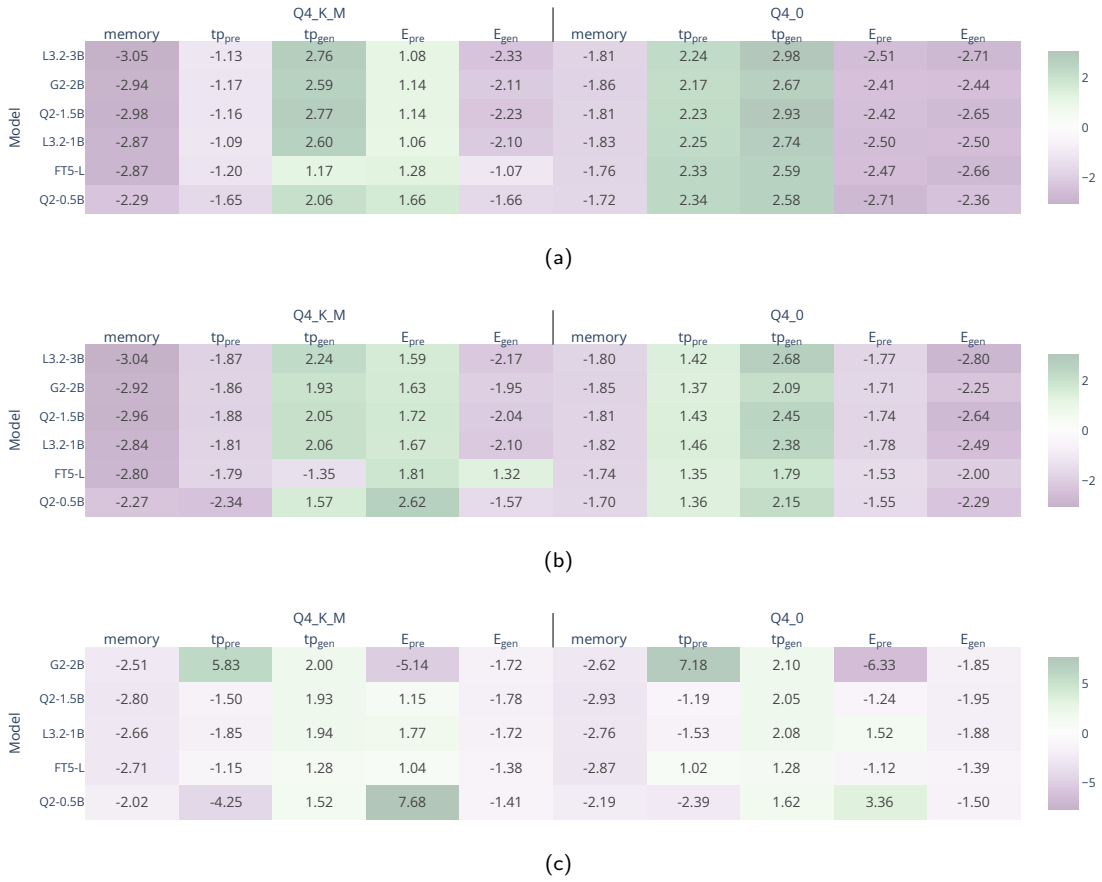


Fig. 12. Change factors on various metrics on RPi 5 (a), Orin CPU (b), and Orin GPU (c) compared to 16 bit floating point models. Negative numbers indicate decrease by absolute value.

To evaluate the impact of quantization on computational performance, Figure 12 illustrate how quantization affected memory footprint, token throughput, and energy consumption for all models that were small enough